

Group Project Initial Results

Alex Caramanico
Wooje Chang
Benjamin Knight
Bahereh Vojdani

**Spring
2025**

User Group

Facility Managers at educational institutions

- Professional expertise in building management
- Need to make data-driven decisions to improve energy efficiency
- Understand technical terminology related to building systems
- Require detailed analysis to prioritize renovation projects
- Budget responsibilities, need to justify investments for building improvements - supported by data

Visual aide will make useful transformations and displaying of data to help facility managers make decisions more quickly and effectively, rather than searching and sorting through considerable data quantities by hand.

Primary Goal and Tasks

High-Level Goal:

**Optimize energy efficiency
by identifying buildings
that require renovation
and/or energy
conservation measures**

Low-Level Tasks

1. Identify patterns in *energy consumption* across different *building types* (residential vs. office vs. lab) to determine if certain building categories consistently use more energy than others
2. Analyze the relationship between a *building's age* and its *energy performance* to prioritize renovation projects
3. Compare *energy usage intensity* (energy use per square foot) across buildings to identify outliers that may need immediate attention
4. Evaluate *seasonal energy consumption patterns* to locate unsatisfactory energy trends
5. Qualify estimated *renovation costs* to develop a value-based approach for building renovations

Dataset

Static Data

- 172 items, 19 attributes (~9 relevant)
- Attributes:
 - Building Name and Type
 - Total CO2 (kkg) and Total Energy (kBtu) from Feb. 2020 to Oct 2024
 - Area (sq. feet)
 - Year Built
 - Annual and Benchmark Energy Usage Intensity (EUI)
 - Renovation Cost (USD) (see appendix)

Time-Series Data

- 176 items, 60 time points → 10,032 items, 8 attributes (see appendix)
- Attributes (after reformat):
 - Building Name and Type
 - Total and *monthly* Energy (kBtu)
 - Month and Year
 - Area (sq. feet)
 - Year Built
 - Annual and Benchmark EUI

Sample data items

	Building Name	Building ID	Total (kkg CO ₂ e)	Total (kBtu)_Total Energy	Area(sq ft)	Year Built	UMD Type	Annual EUI	EPA Benchmark EUI	Renovation_Cost
1	A James Clark Hall	429	8873769	212533234	182440	2017	Lab	252	77	4173824
2	A V Williams Building	115	3539686	200694928	236044	1988	Office	182	69	145500494
3	Agriculture Shed	102	0	714284	2322	1938	Lab	74	47	477054
4	Allegany Hall	24	418315	9502466	40300	1955	Residential	50	64	22475577
5	Alpha Chi Omega Sorority (4525 College Ave)	172	157247	6410615	11712	1960	Residential	121	56	894775
6	Alpha Delta Pi Sorority (4535 College Ave)	170	116683	5105839	10459	1959	Residential	110	55	799048
7	Alpha Phi Sorority (7402 Princeton Ave)	176	52696	5552399	11833	1964	Residential	103	55	6599343
8	Animal Sciences Agric Engrg Bldg	142	4073235	213804143	195450	1970	Lab	203	72	120477841
9	Annapolis Hall	8	237235	7338030	22855	1988	Office	65	59	11538833
10	Anne Arundel Hall	60	0	5564543	36651	1937	Residential	44	61	20440506

	Building Name	Month	Year	Monthly Energy	Total (kBtu)	Area(sq ft)	Year Built	UMD Type	Annual EUI	EPA Benchmark EUI
177	Agriculture Shed	Feb	2020	16597	714284	2322	1938	Lab	74	47
178	A V Williams Building	Feb	2020	3493572	200694928	236044	1988	Office	182	69
179	Chestertown Hall	Feb	2020	291256	7478306	29090	1962	Residential	45	59
180	Cumberland Hall	Feb	2020	865680	29123716	124486	1963	Residential	49	64
181	House 1	Feb	2020	31941	3727356	10594	1958	Residential	75	54
182	House 2	Feb	2020	35623	3049145	9144	1958	Residential	75	53
183	House 3	Feb	2020	20988	3108969	9464	1954	Residential	67	52
184	House 4	Feb	2020	38105	1979406	9464	1954	Residential	43	52
185	House 5	Feb	2020	19113	2768111	9464	1954	Residential	51	55
186	House 6	Feb	2020	26954	2581886	9464	1954	Residential	66	52

Memory Management We will most likely store the data in browser memory, since it is not a prohibitively large amount of data. 4

Tableau Visualizations

Tasks 1 and 4 - Box and Whisker

Attributes: Building Type, Year, Total Monthly Energy (each building)

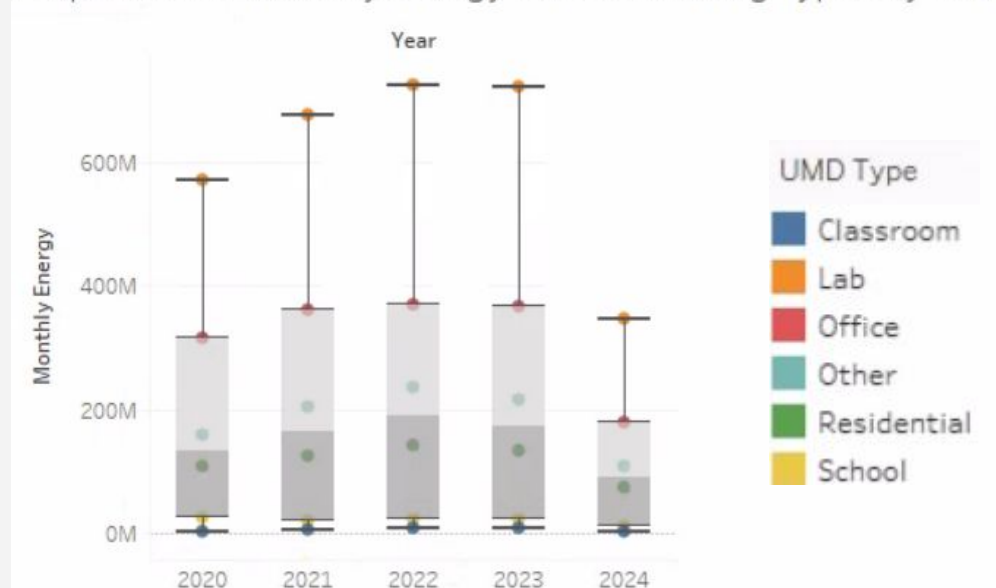
Transformations: None

Effectiveness: 2.5

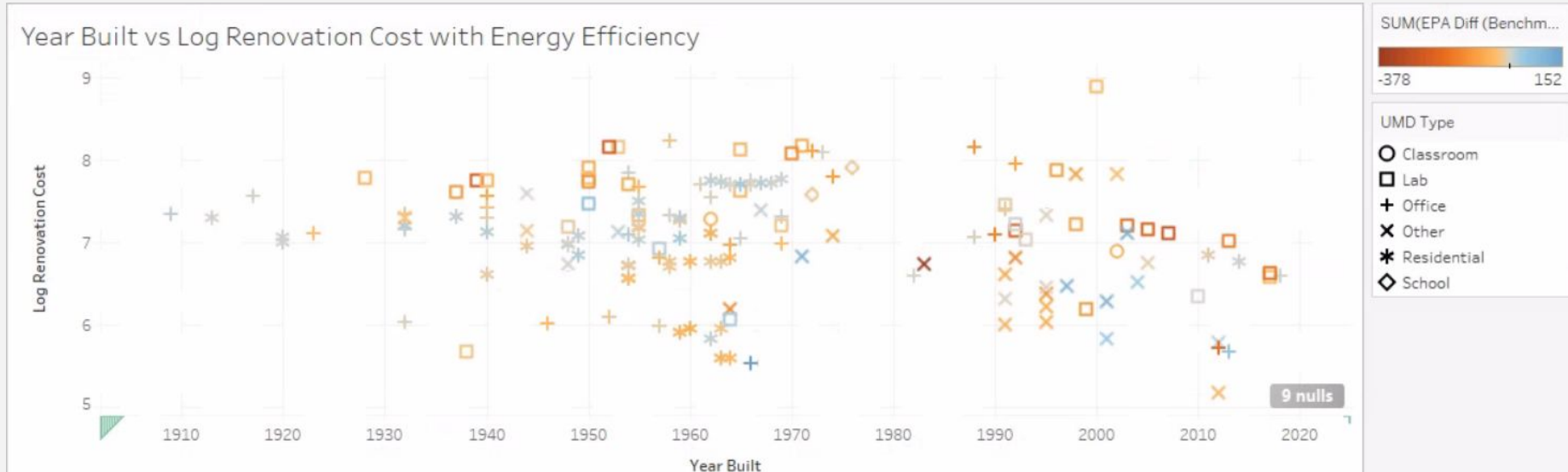
Pros and Cons:

This visualization simply conveys the quartiles of average monthly energy per building across each building type. It is effective for comparisons between the categories through time. However, the quartiles are not as meaningful with only six values per year, and it is hard for the user to understand that the quartiles are between aggregates of building type. A more effective example for the time series data can be found on slide 11.

Dispersion of Monthly Energy Across Building Types By Year



Tasks 1, 2, 3, 5 - Scatter Plot



Attributes: Building Type, Year Built, Log Renovation Cost, EUI Difference, Building Name

Transformations:

EUI Difference = EPA Benchmark EUI - Annual EUI

Log Renovation Cost = $\log_{10}(\text{Renovation Cost})$

Effectiveness: 3.75

Pros and Cons:

This visualization takes advantage of shapes and colors to show both the value of emissions and building categories. The color scale may pose accessibility concerns, and building labels could be helpful. It may also be good to clean up the clutter, perhaps with filtering mechanisms. However, the task relates to four of the five tasks, improving its utility.

Tasks 1, 2, and 3 - Stacked Bar Chart

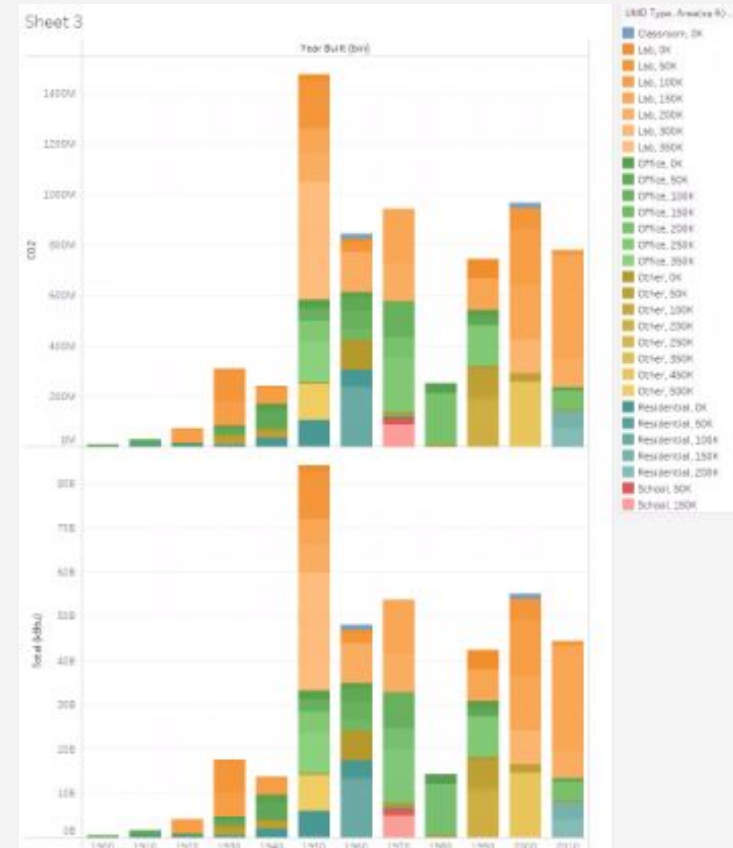
Attributes: Building Type (divided into separate groups by building area), Year Built, Total CO₂, Total Energy

Transformations: None

Effectiveness: 3

Pros and Cons:

This visualization is effective in encoding several attributes - one can analyze carbon dioxide or energy consumption, and make comparisons between the two. Also, these variables can be contextualized with construction date (Task 2). The bar sizes encode the relative contribution of each building type, which are further subdivided into buckets across building sizes. This gives insight to Task 3 - energy efficiency (a function of area). There are some setbacks with this plot though, firstly that it is difficult to discern the actual cumulative values of each sub-group in a stacked bar chart. There may also be too many groups, making it difficult to discern between them. It also does not have building-level detail, but this could be added as a drill-down feature. Lastly, it is unclear how/why buildings were grouped together in the way they were for area and year built.



Additional Visualizations (not creatable with Show Me)

Task 1 - Layered Circle (Python)

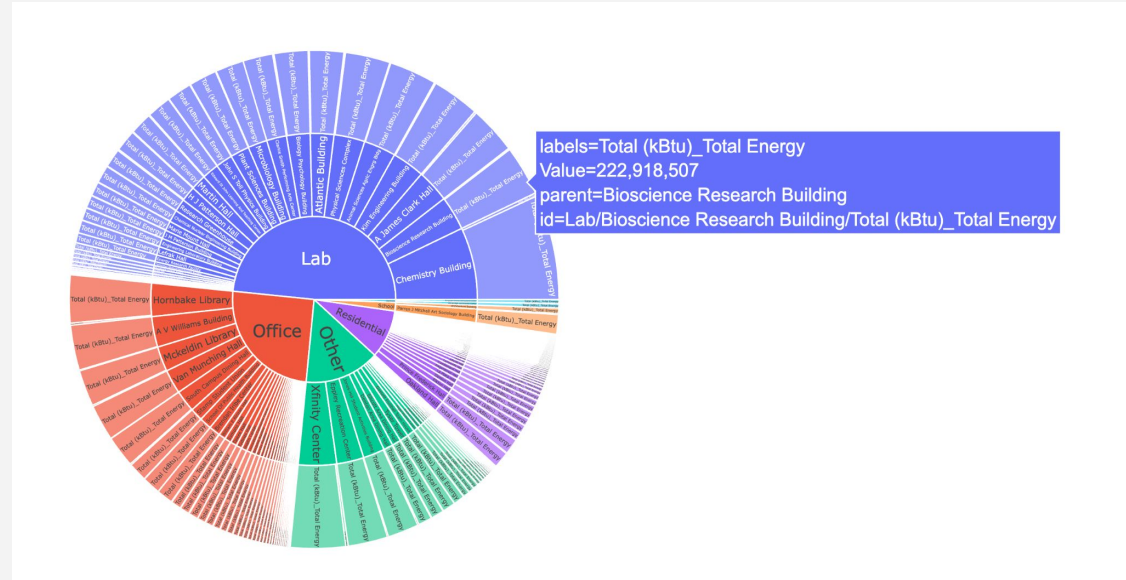
Attributes: Building Type, Building Name, Total Energy (kBtu)

Transformations: None

Effectiveness: 4.0

Pros and Cons:

This visualization is highly flexible and customizable. It is also very intuitive - the user can quickly understand how area corresponds to total energy emissions. However, its cluttered initial appearance may make it difficult for users to decipher, and it is limited to showing only one quantitative attribute in its current state.



Note: this visualization has interactive (zooming) functionality - see attached file. This is a pro, but could also be a con as interaction requires “an extra learning step” to gain specific insights.

Tasks 1, 3, and 4 - Line Charts

Attributes: Building Type (Lines on top, Pages on bottom), Month and Year, EUI Difference, Building Name (lines on bottom), Average Monthly Energy (across buildings or individually)

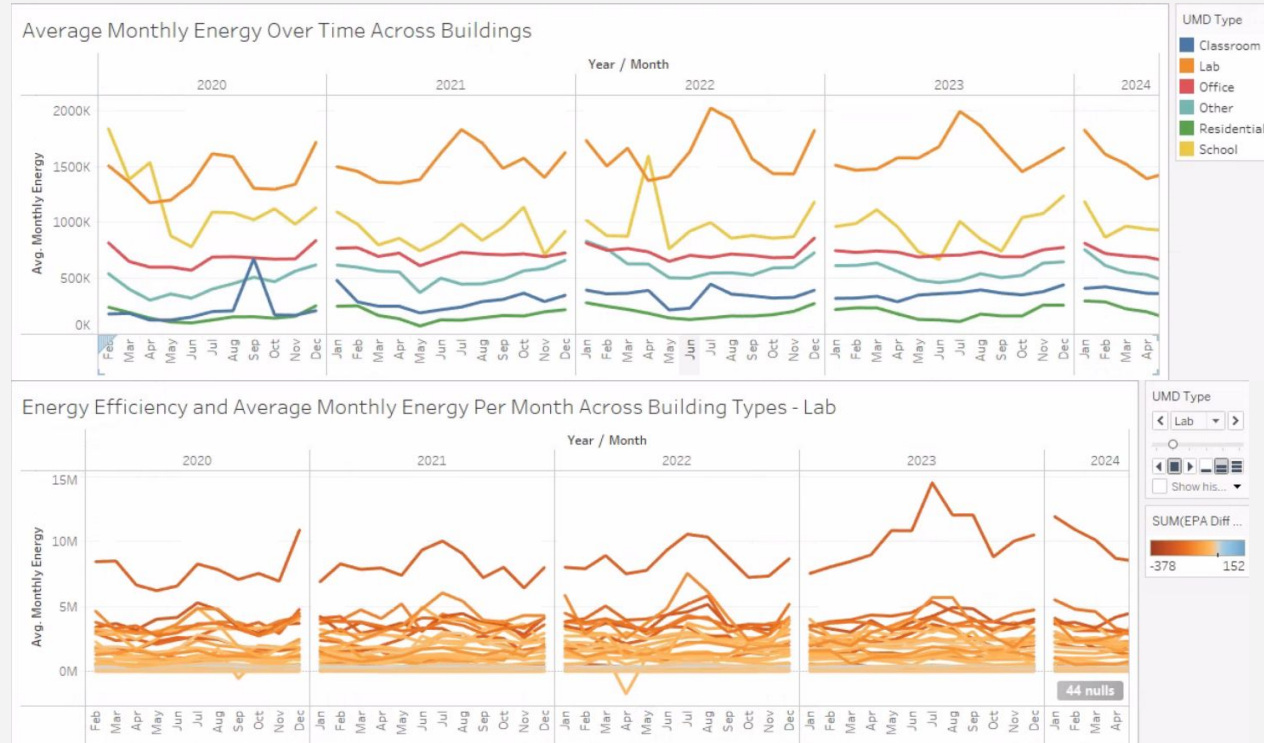
Transformations:

EUI Difference = EPA Benchmark EUI - Annual EUI

Effectiveness: 4.25

Pros and Cons:

This communicates emission trends by giving both a broader categorical aggregate and building specific values. It gives insight into trends, differences between building types, and relative energy efficiency. The user can also hover over a line to see the building it represents. The main downside is the “business” of the plot, as many lines overlap, making comparisons difficult.



Appendix

The original time-series data was not formatted properly for Tableau, and was thus altered in the following way using R:

1. Time-series data (bottom left) joined with static data on Building Name, selecting variables of interest from latter data set
2. Resulting data was pivoted to a long format, with the time points split into separate variables for month and year, giving the bottom right result

Building ID	Building Name	Feb 2020 (kBtu)	Mar 2020 (kBtu)	Apr 2020 (kBtu)	May 2020 (kBtu)
1	102 Agriculture Shed	16597	6824	10837	1501
2	115 A V Williams Building	3493572	3086283	3272814	2968026
3	121 Chestertown Hall	291256	234564	202115	126838
4	122 Cumberland Hall	865680	594543	396261	117387
5	126 House 1	31941	119179	39771	27364
6	127 House 2	35623	78180	24551	23530
7	128 House 3	20988	82051	23223	26469
8	129 House 4	38105	94395	26951	19795
9	130 House 5	19113	49089	15670	16209
10	131 House 6	26954	98746	32210	19698



Building Name	Month	Year	Monthly Energy	Total (kBtu)	Area(sq ft)	Year Built	UMD Type	Annual EUI	EPA Benchmark EUI
177 Agriculture Shed	Feb	2020	16597	714284	2322	1938	Lab	74	47
178 A V Williams Building	Feb	2020	3493572	200694928	236044	1988	Office	182	69
179 Chestertown Hall	Feb	2020	291256	7478306	29090	1962	Residential	45	59
180 Cumberland Hall	Feb	2020	865680	29123716	124486	1963	Residential	49	64
181 House 1	Feb	2020	31941	3727356	10594	1958	Residential	75	54
182 House 2	Feb	2020	35623	3049145	9144	1958	Residential	75	53
183 House 3	Feb	2020	20988	3108969	9464	1954	Residential	67	52
184 House 4	Feb	2020	38105	1979406	9464	1954	Residential	43	52
185 House 5	Feb	2020	19113	2768111	9464	1954	Residential	51	55
186 House 6	Feb	2020	26954	2581886	9464	1954	Residential	66	52

Renovation Cost Estimates: <https://facilities.umd.edu/info-resources/building-inventory>

Extra Data Resources: fmwebmaster@umd.edu, sustainabilitydata@umd.edu

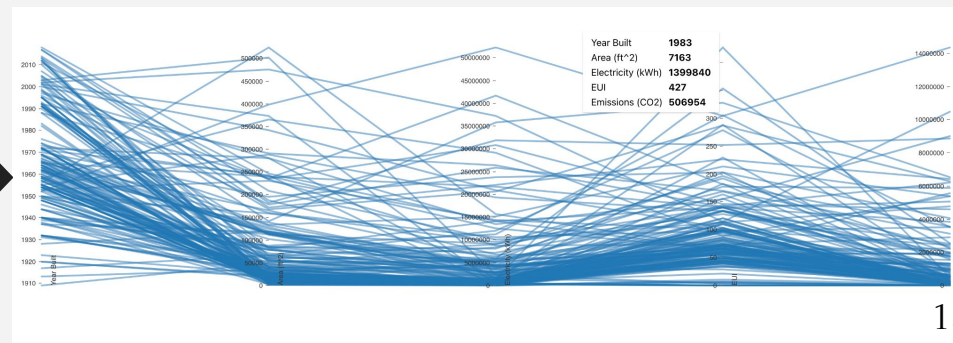
Bonus Visualizations

We wanted to include some additional visualizations that we considered including in the presentation. Most are still different types to those shown before, but were not significantly different in terms of attribute selection or interpretation. We decided to include them here in the hopes of getting feedback on them as well, if possible. Slide 15 was originally in our selected 5, but we felt it was too similar to Slide 7 in terms of chart type. It is still included as we agreed it is very effective. Slide 7 was kept for its use of Renovation Cost (Task 5)

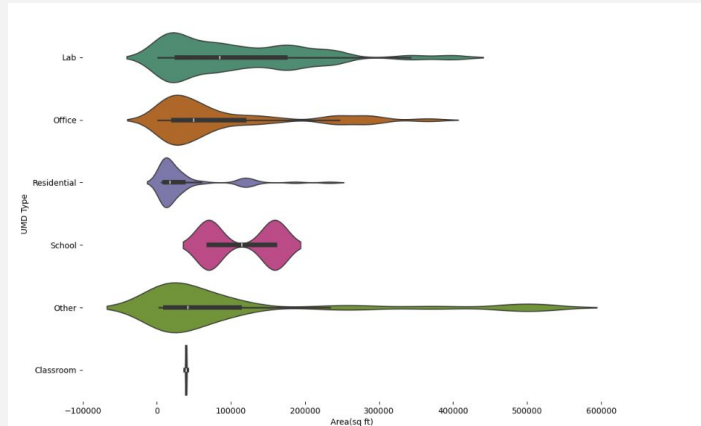
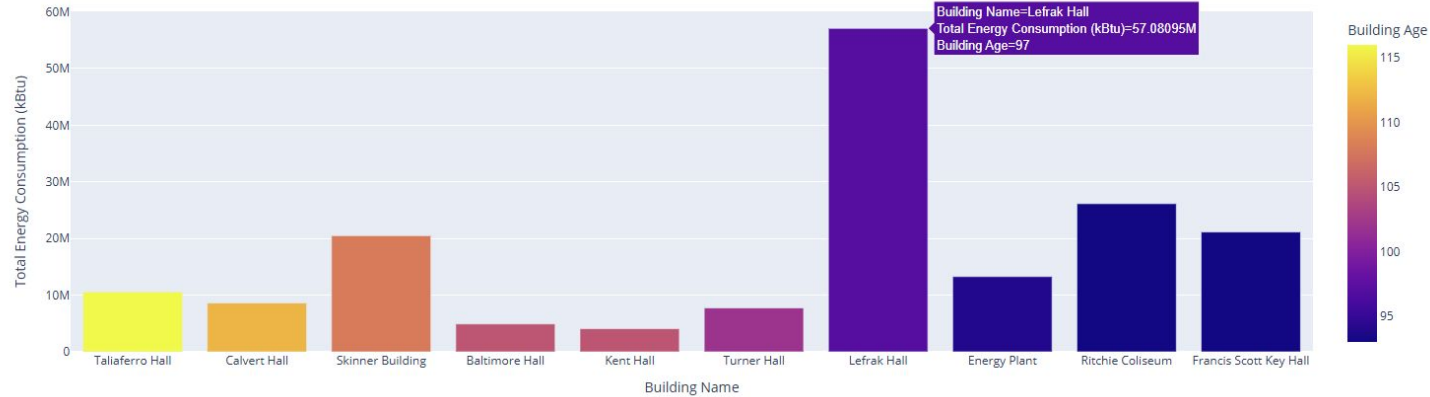


Tasks 1 and 4
(Tableau)

Tasks 1,2,3,5
(potential)
(JS)

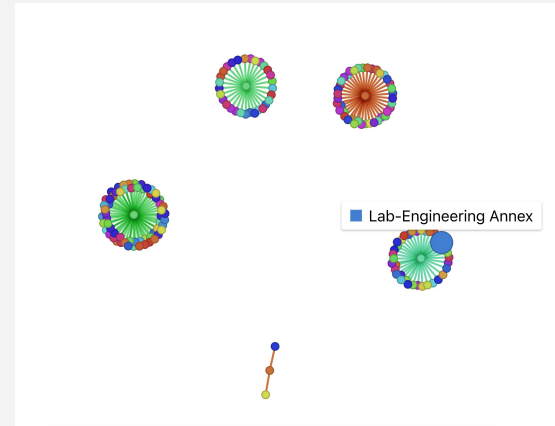


Top 10 Buildings for Renovation Priority



Task 2
(JS)

Task 1
(JS)



Tasks 1, 2, and 3 - Categorical Plot

Attributes: Building Type, Year Built, EUI Difference, Building Name (points)

Transformations: EUI Difference = EPA Benchmark EUI - Annual EUI

Effectiveness: 4.75

Pros and Cons:

This visualization is primarily focused on conveying which periods of time saw the most carbon intensive buildings built. It also shows the relative EUI performance of each building. This functions as a simpler derivative of the second (slide 7) visualization, streamlining information by excluding renovation costs. However, without building labels, it is hard to make further comparisons. Additionally, dots tend to be stacked on top of each other, making it difficult to distinguish emissions intensity.

